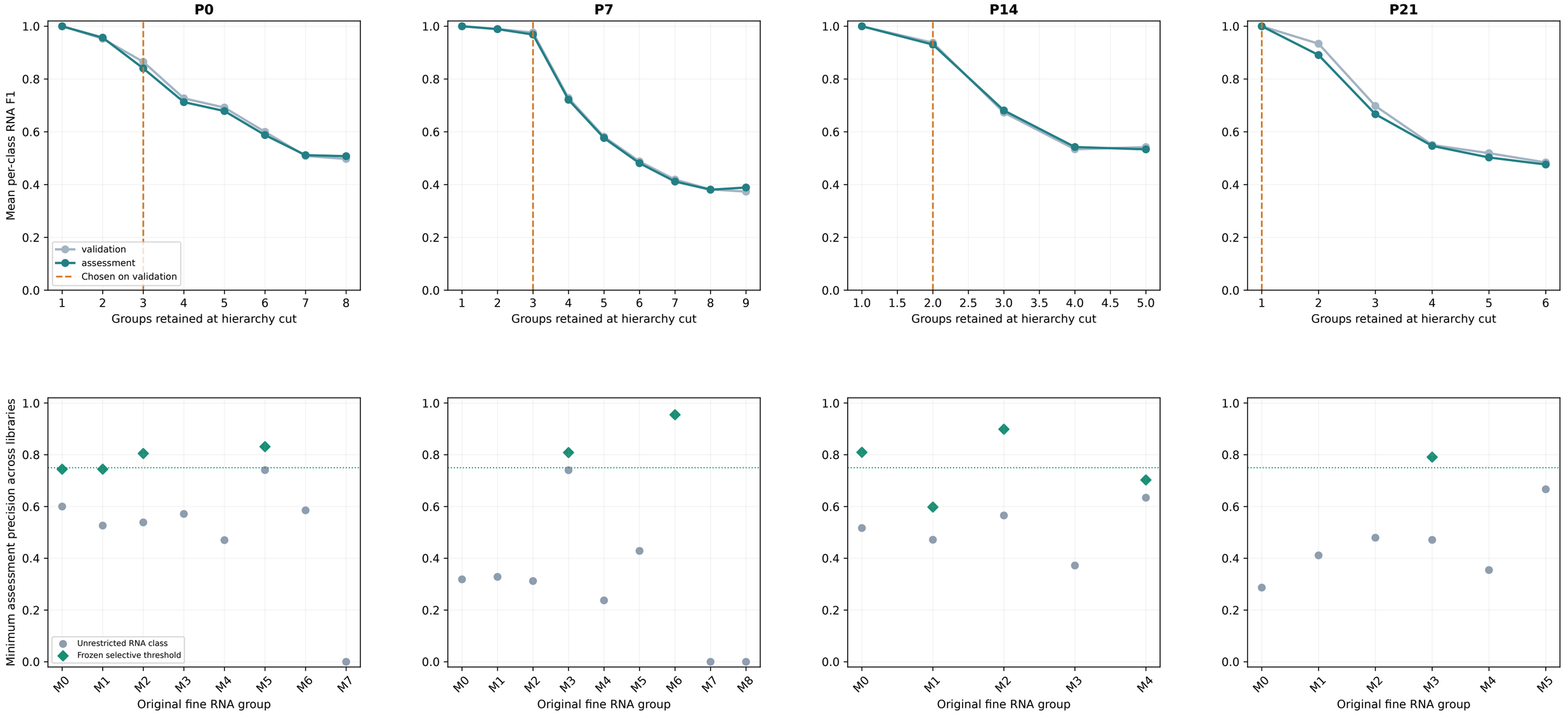
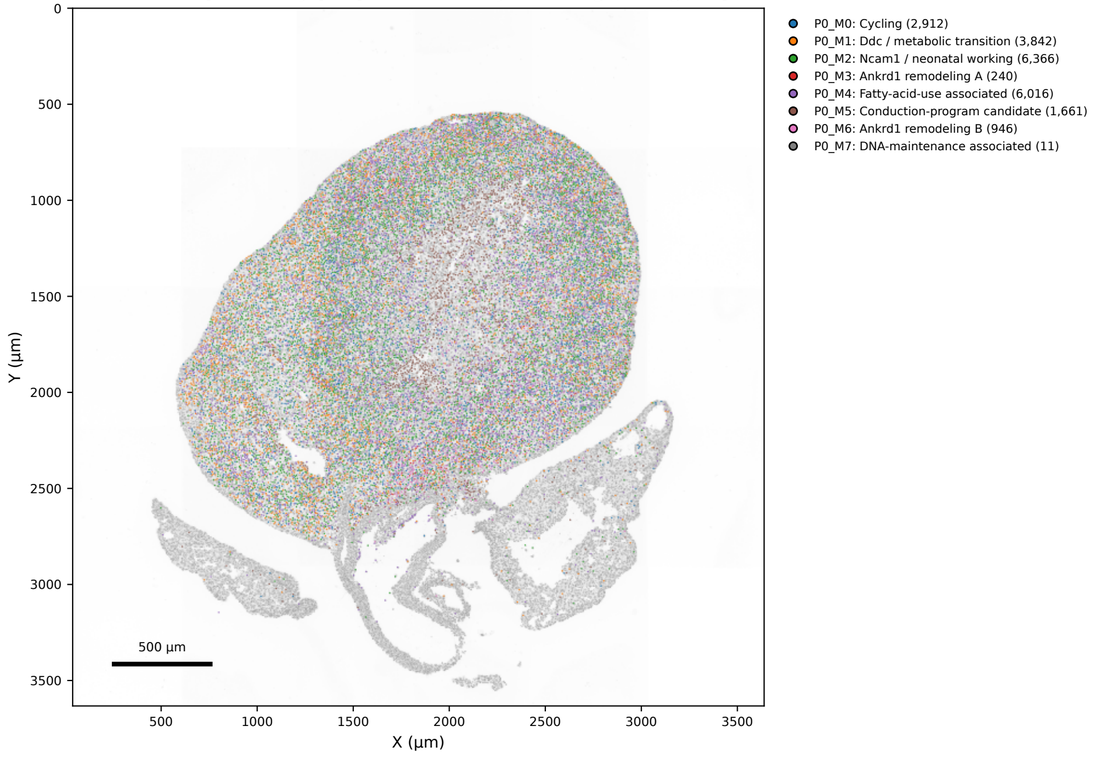
How far can the shared panel distinguish the existing ventricular RNA groups?



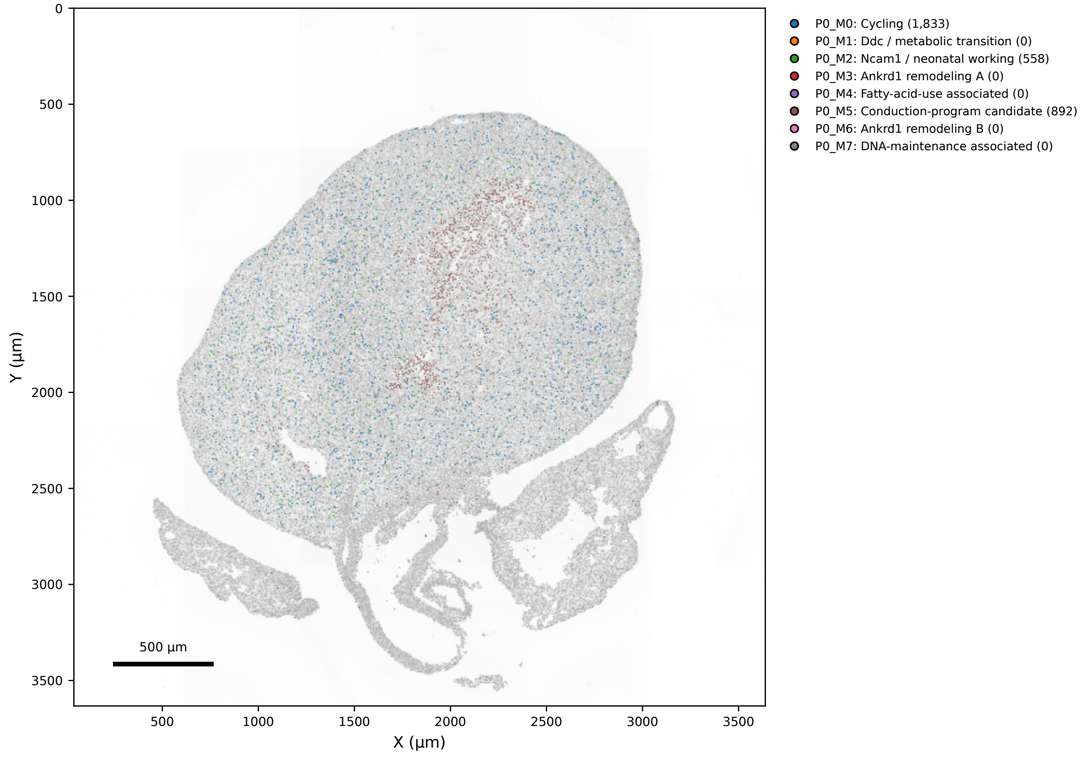
Hierarchy merges use calibration nuclei; cut selection uses validation nuclei; the assessment quarter evaluates the frozen choices.
Bottom: selective precision is shown only where a threshold was chosen on validation. These are internal RNA-label tests, not Xenium anatomical accuracy.

P0: pushing ventricular spatial mapping to 8 existing RNA groups

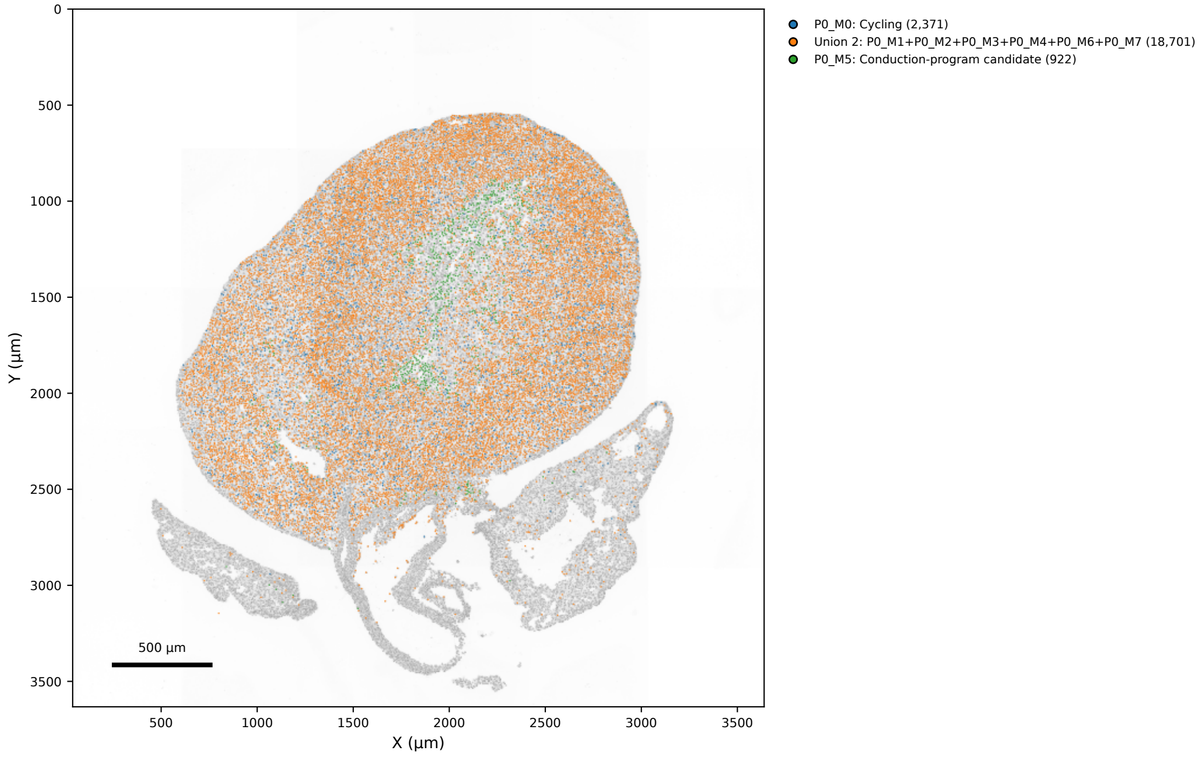
Finest exploratory proposals: every existing RNA group



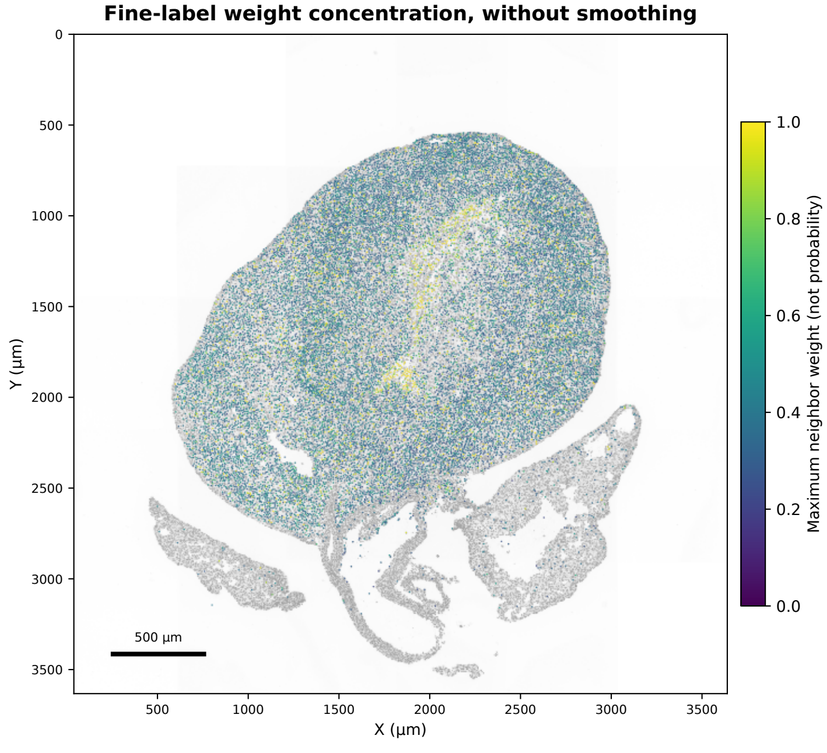
Fine candidates with either internal support check



Coarser cut selected on reserved RNA validation nuclei



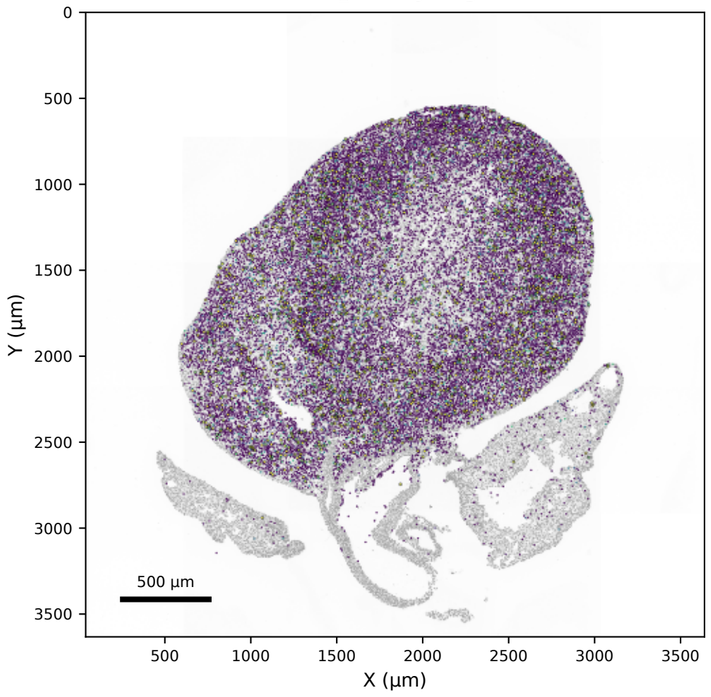
Fine-label weight concentration, without smoothing



Top left is a permissive hypothesis map; top right adds internal RNA support, library/raw consistency and measured-marker co-detection.
The merged unions are computational resolution checks, not newly discovered biological cell types. No coordinates are moved or labels spatially smoothed.
Conduction labels remain candidates. Kcnd2/Irx5 are unassayed; all working/electrical labels originate from the RNA reference.

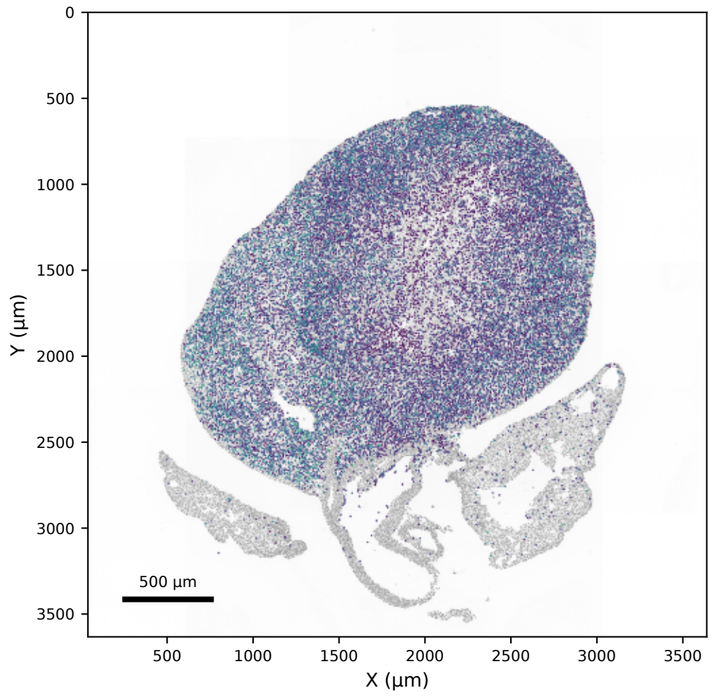
P0: individual spatial evidence maps for all 8 ventricular RNA groups

P0_M0 · Cycling



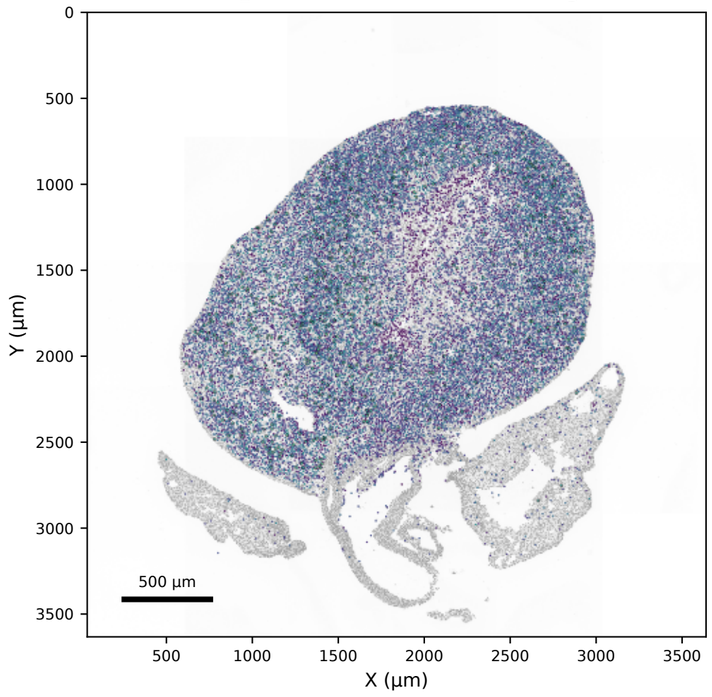
Proposed: 2,912; prioritized: 1,833
Measured supporting genes: Atad2, Top2a, Rad51b

P0_M1 · Ddc / metabolic transition



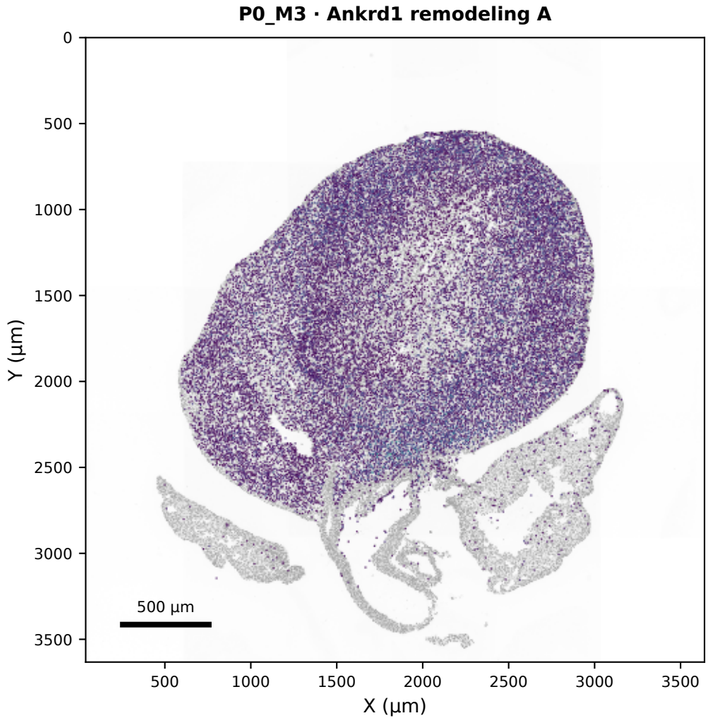
Proposed: 3,842; prioritized: 0
Measured supporting genes: Cacna1d, Angpt1, Ivns1abp

P0_M2 · Ncam1 / neonatal working



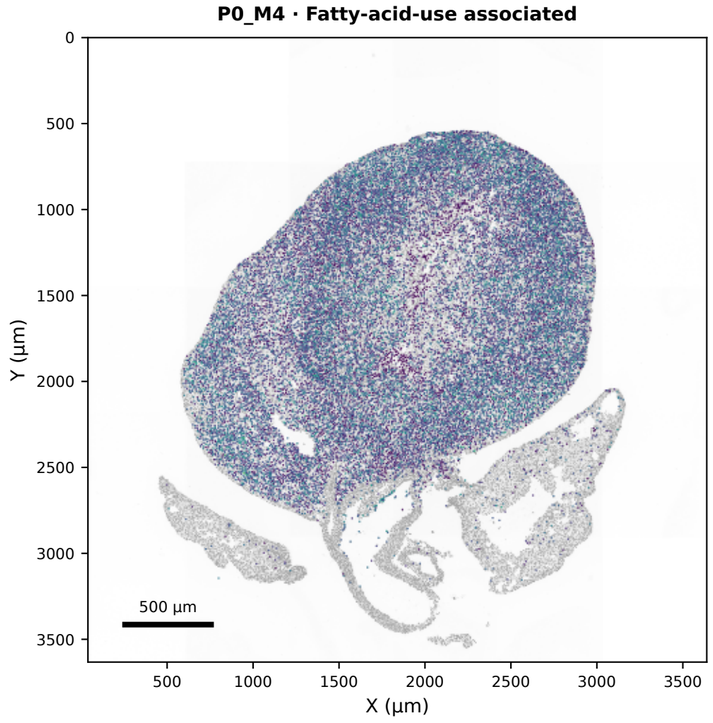
Proposed: 6,366; prioritized: 558
Measured supporting genes: Ncam1, Mapk10, Kcnq5

P0_M3 · Ankrd1 remodeling A



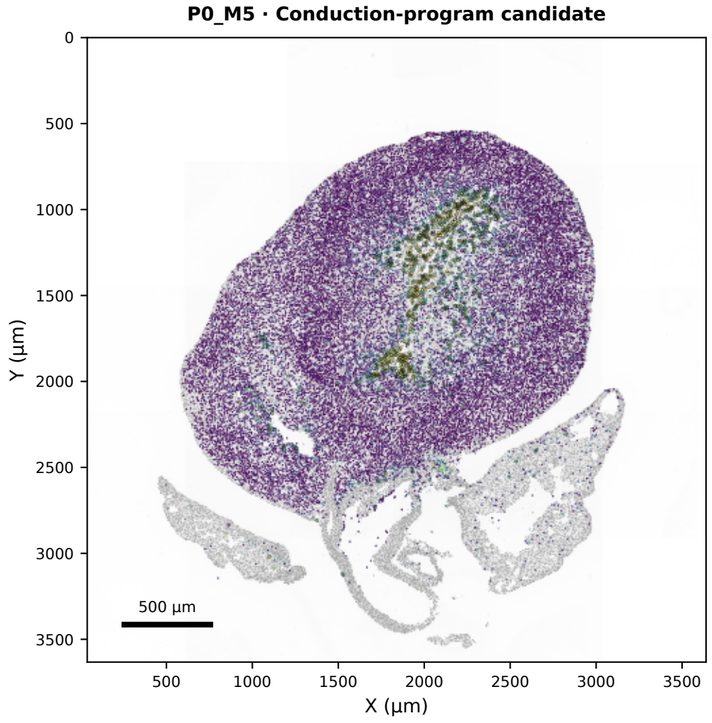
Proposed: 240; prioritized: 0
Measured supporting genes: Ankrd1, Creb5, Itga9

P0_M4 · Fatty-acid-use associated



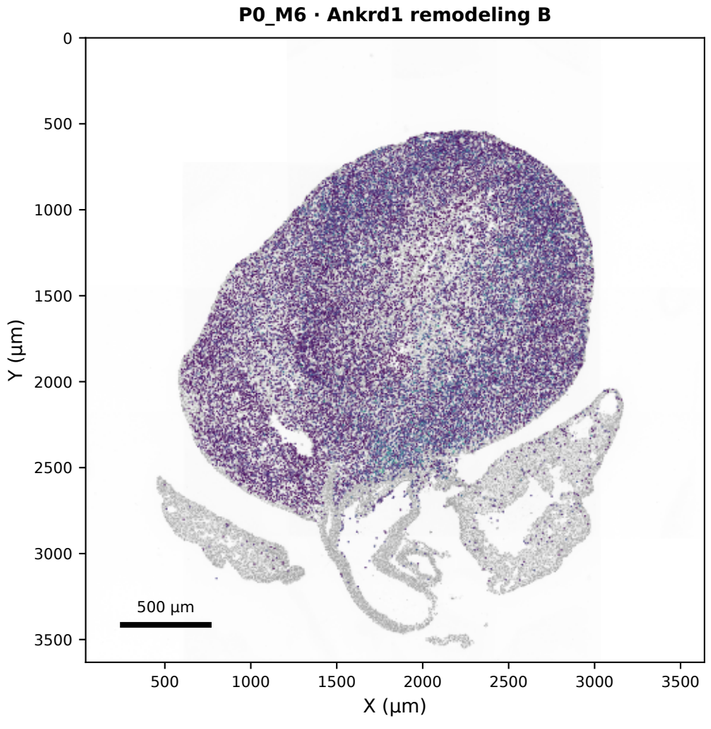
Proposed: 6,016; prioritized: 0
Measured supporting genes: Eda, Dnmt3a, Cd36

P0_M5 · Conduction-program candidate



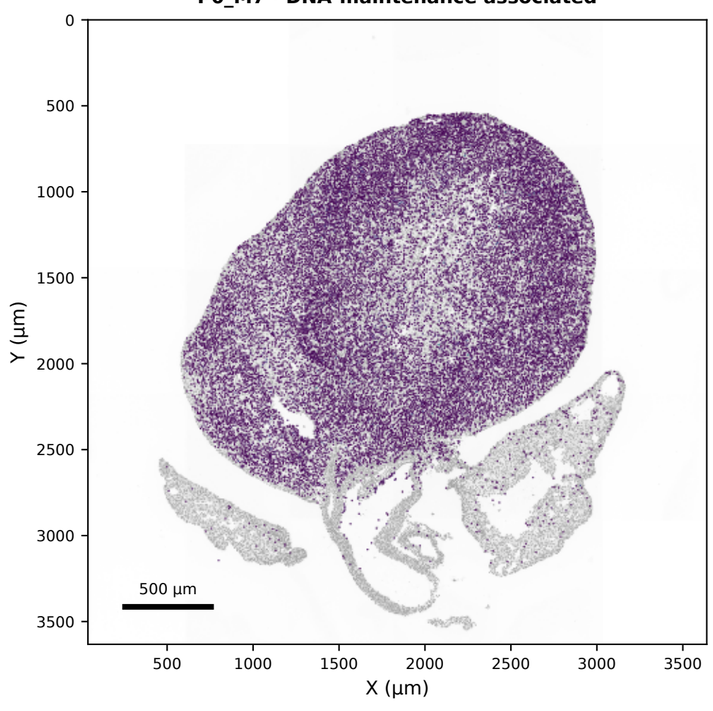
Proposed: 1,661; prioritized: 892
Measured supporting genes: Slit2, Sema3a, Thsd4

P0_M6 · Ankrd1 remodeling B

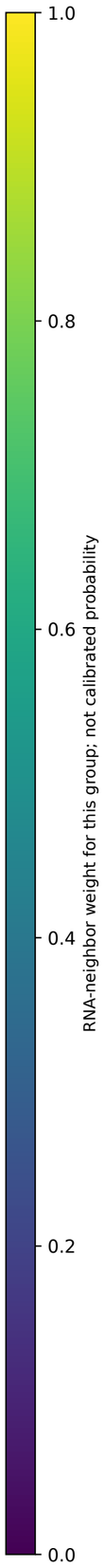


Proposed: 946; prioritized: 0
Measured supporting genes: Ankrd1, Creb5, Acta2

P0_M7 · DNA-maintenance associated

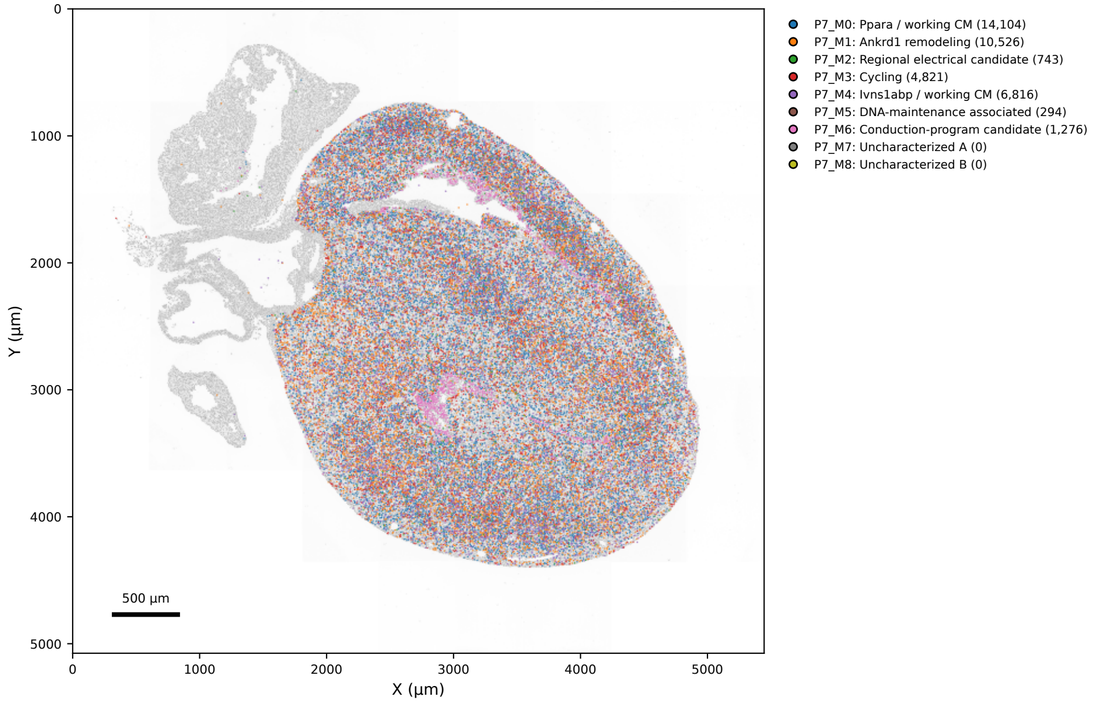


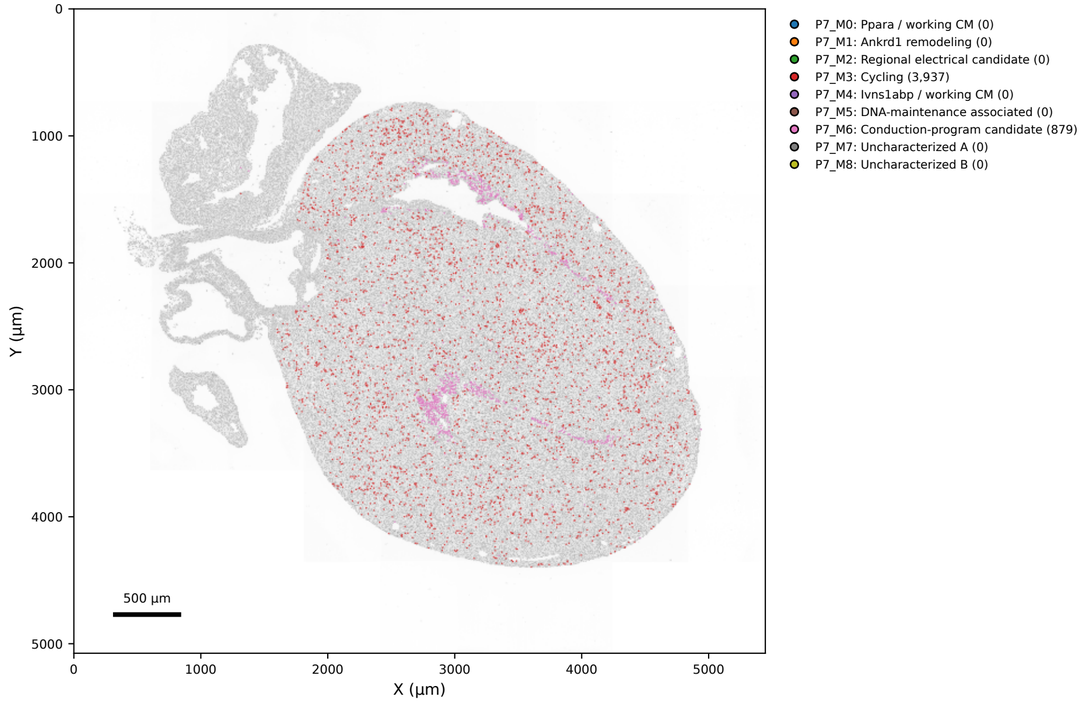
Proposed: 11; prioritized: 0
Measured supporting genes: Pola1, Rad51b, Hells

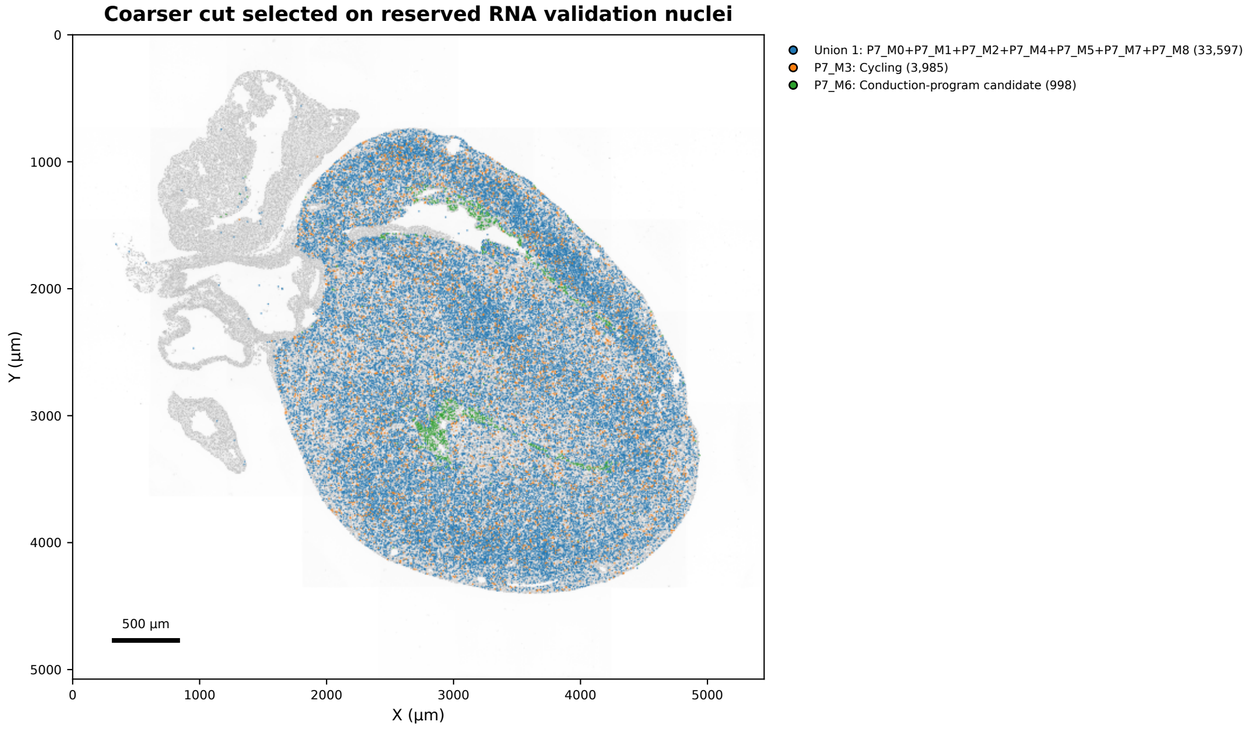


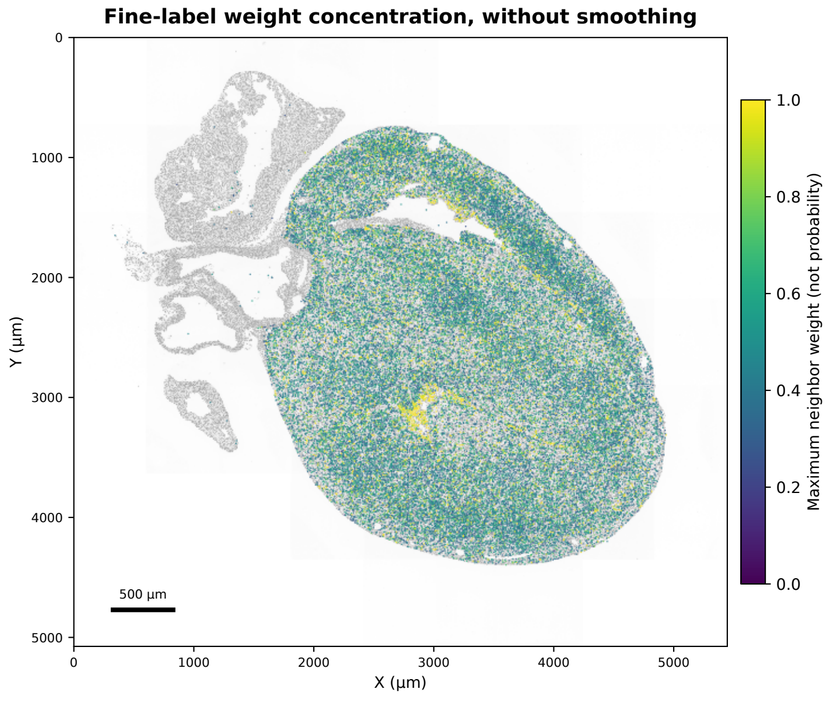
All original locations are retained. Colors show continuous group weight; black outlines mark prioritized candidate placements.
Measured markers corroborate labels but also contributed to matching. Low weights and absent outlines do not establish biological absence.

P7: pushing ventricular spatial mapping to 9 existing RNA groups





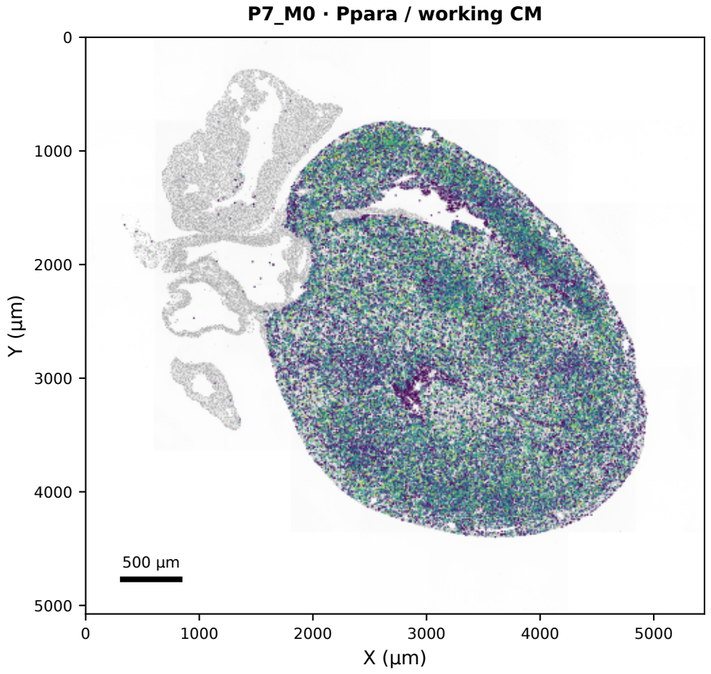




Top left is a permissive hypothesis map; top right adds internal RNA support, library/raw consistency and measured-marker co-detection.
The merged unions are computational resolution checks, not newly discovered biological cell types. No coordinates are moved or labels spatially smoothed.
Conduction labels remain candidates. Kcnd2/Irx5 are unassayed; all working/electrical labels originate from the RNA reference.

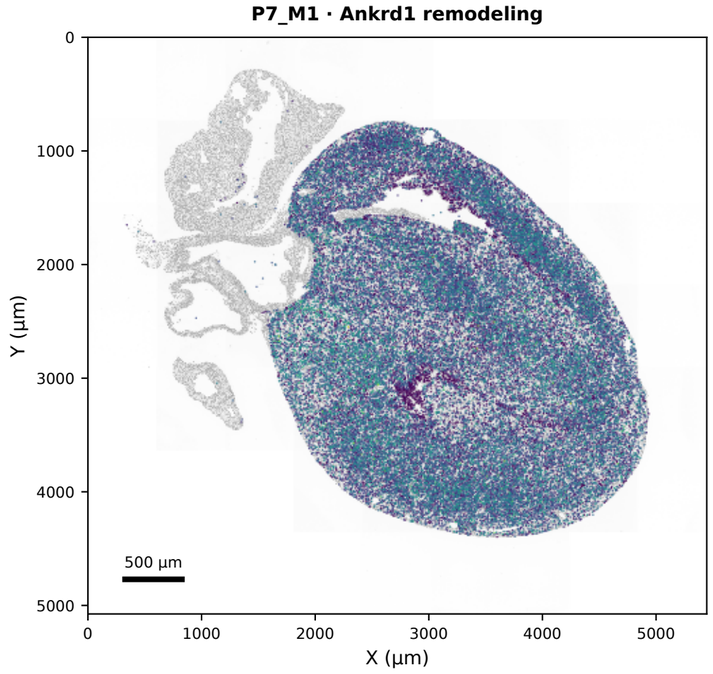
P7: individual spatial evidence maps for all 9 ventricular RNA groups

P7_M0 · Ppara / working CM



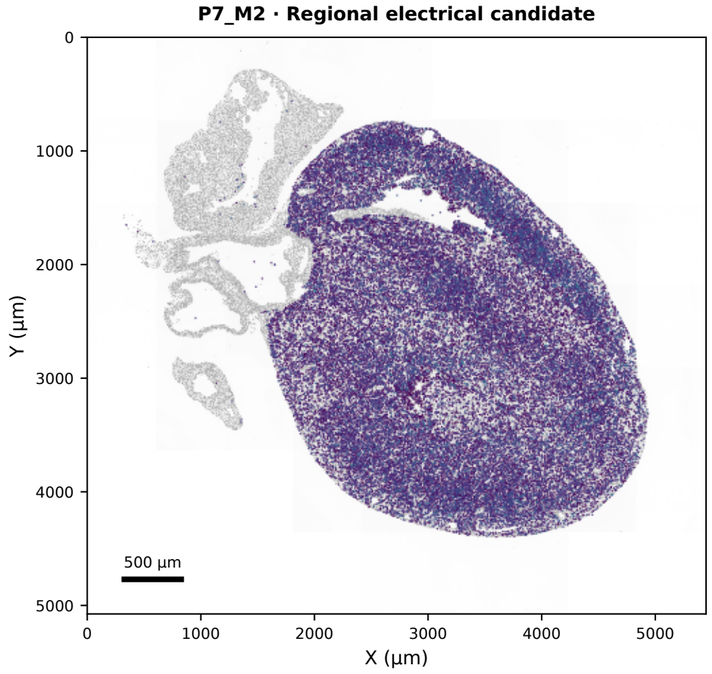
Proposed: 14,104; prioritized: 0
Measured supporting genes: Rreb1, Ppara, Itgb6

P7_M1 · Ankrd1 remodeling



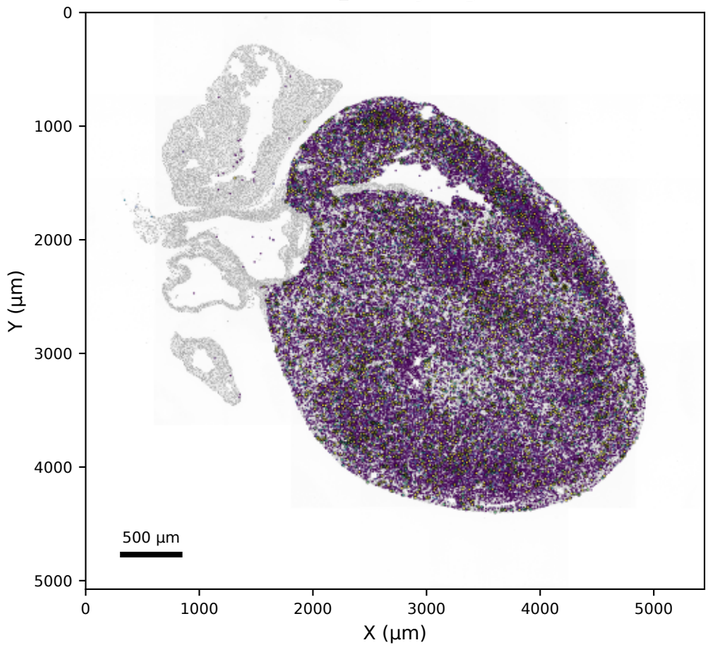
Proposed: 10,526; prioritized: 0
Measured supporting genes: Creb5, Ankrd1, Itga9

P7_M2 · Regional electrical candidate



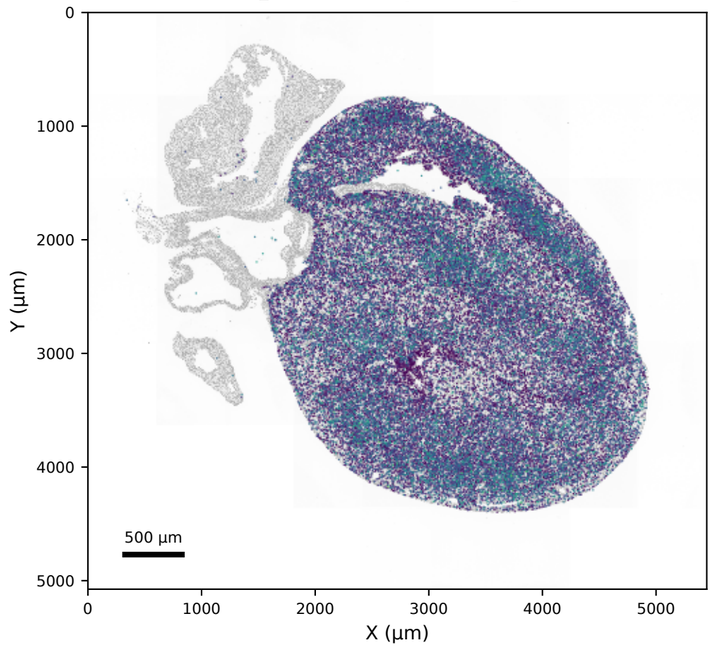
Proposed: 743; prioritized: 0
Measured supporting genes: Thsd4, Gm12381, Angpt1

P7_M3 · Cycling



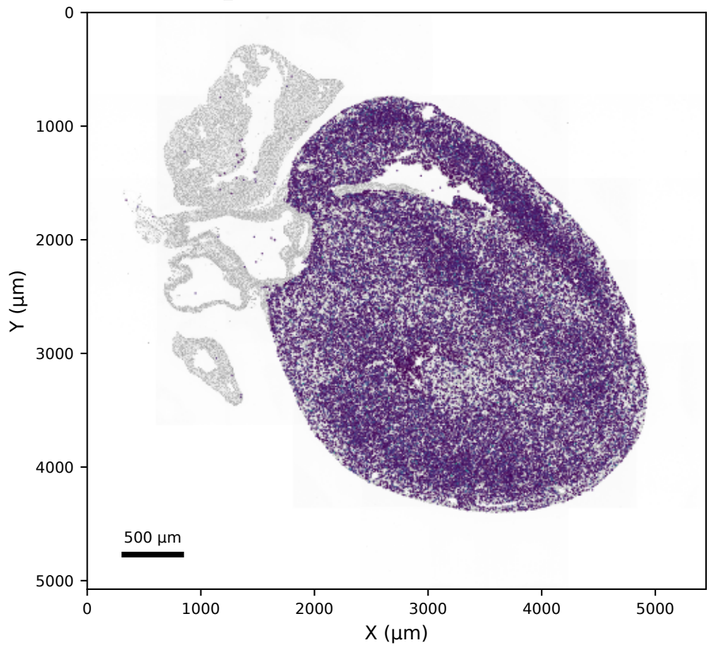
Proposed: 4,821; prioritized: 3,937
Measured supporting genes: Top2a, Mki67, Hjurp

P7_M4 · Ivns1abp / working CM



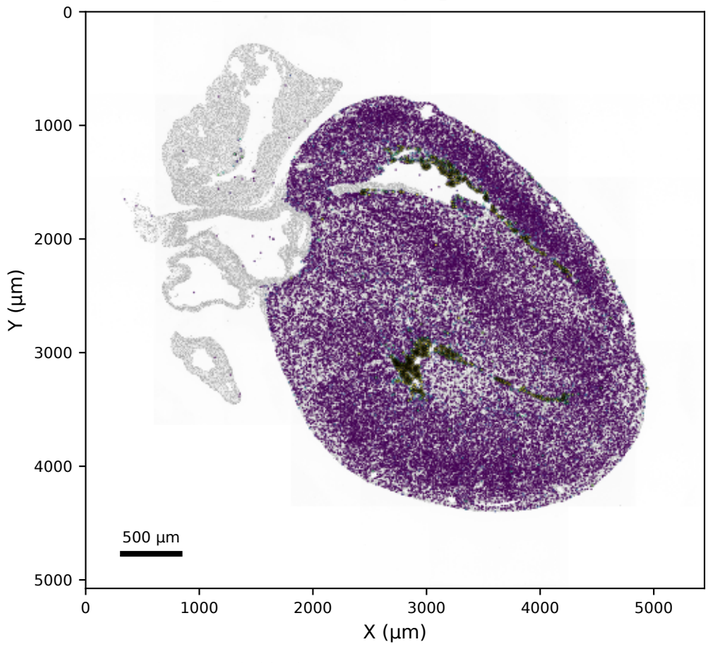
Proposed: 6,816; prioritized: 0
Measured supporting genes: Ivns1abp

P7_M5 · DNA-maintenance associated



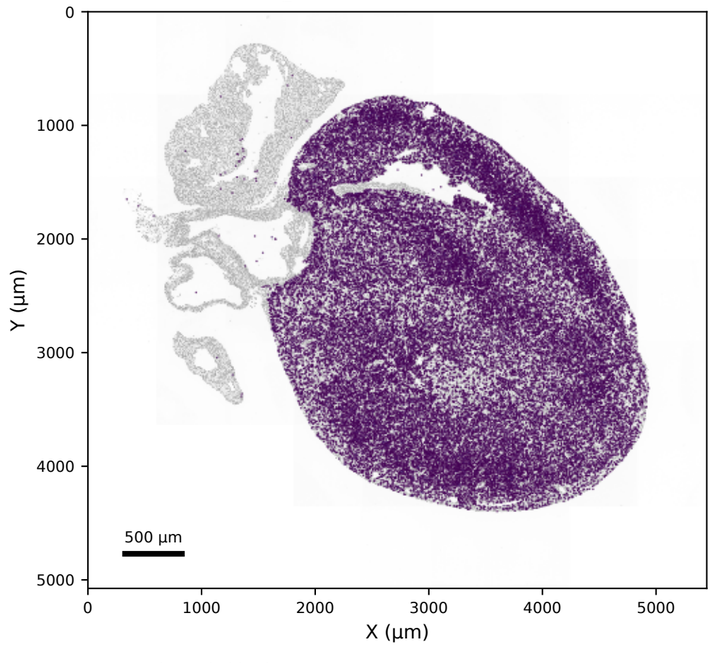
Proposed: 294; prioritized: 0
Measured supporting genes: Pola1, Rad51b, Hells

P7_M6 · Conduction-program candidate



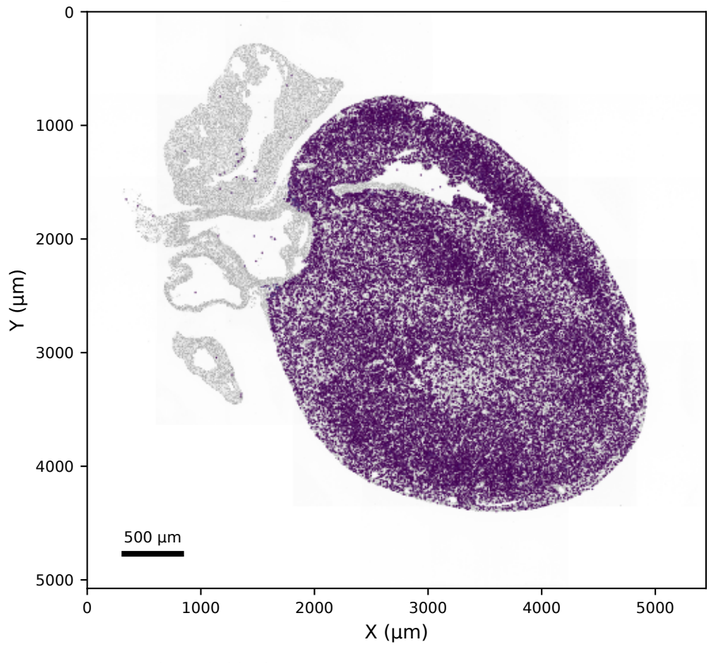
Proposed: 1,276; prioritized: 879
Measured supporting genes: Sema3a, Ntm, Ephb1

P7_M7 · Uncharacterized A

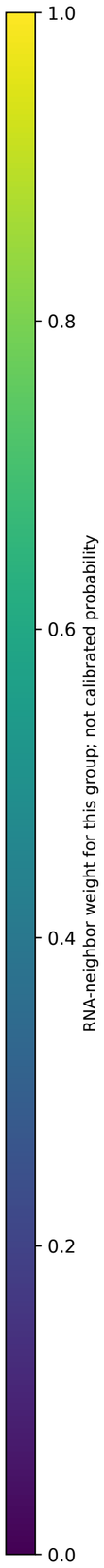


Proposed: 0; prioritized: 0
Measured supporting genes: Cenpf, Sox4, Nrp1

P7_M8 · Uncharacterized B



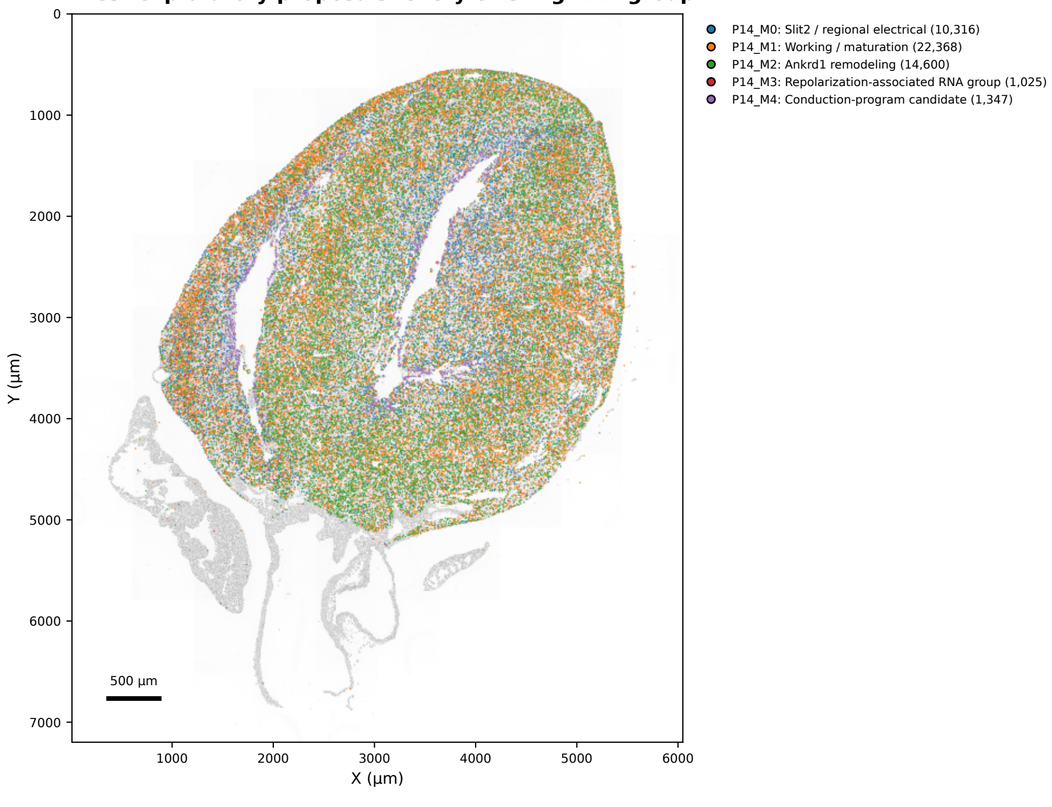
Proposed: 0; prioritized: 0
Measured supporting genes: Cacna1d, Igf1, Sema3a



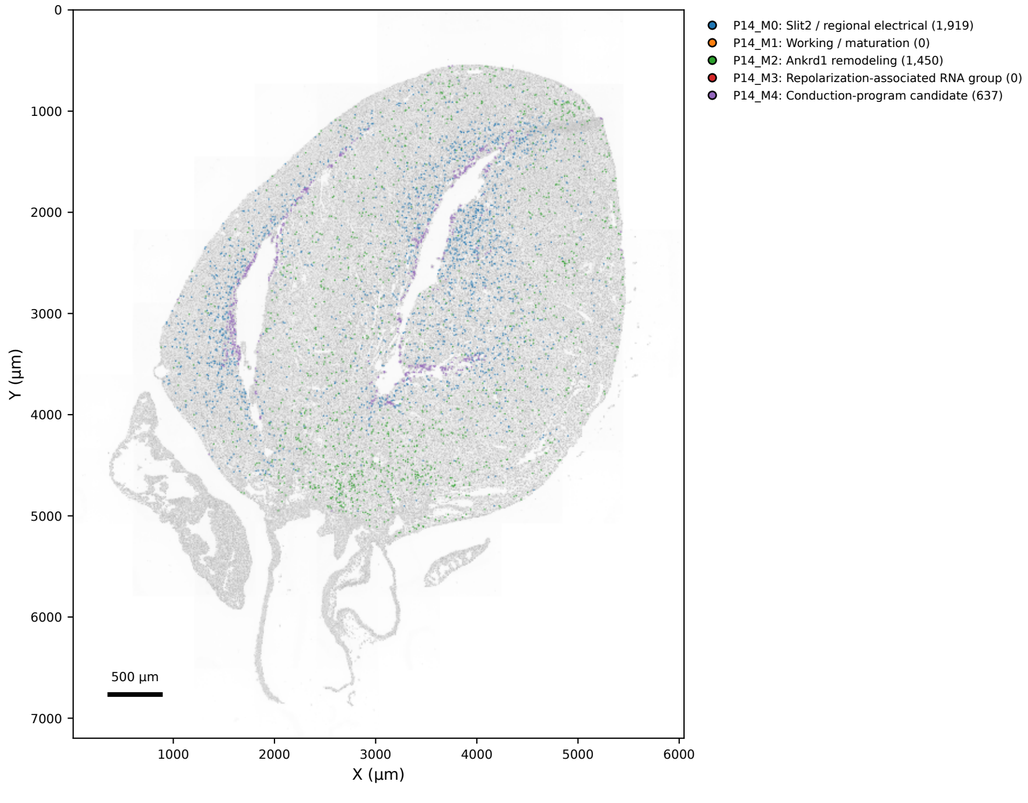
All original locations are retained. Colors show continuous group weight; black outlines mark prioritized candidate placements.
Measured markers corroborate labels but also contributed to matching. Low weights and absent outlines do not establish biological absence.

P14: pushing ventricular spatial mapping to 5 existing RNA groups

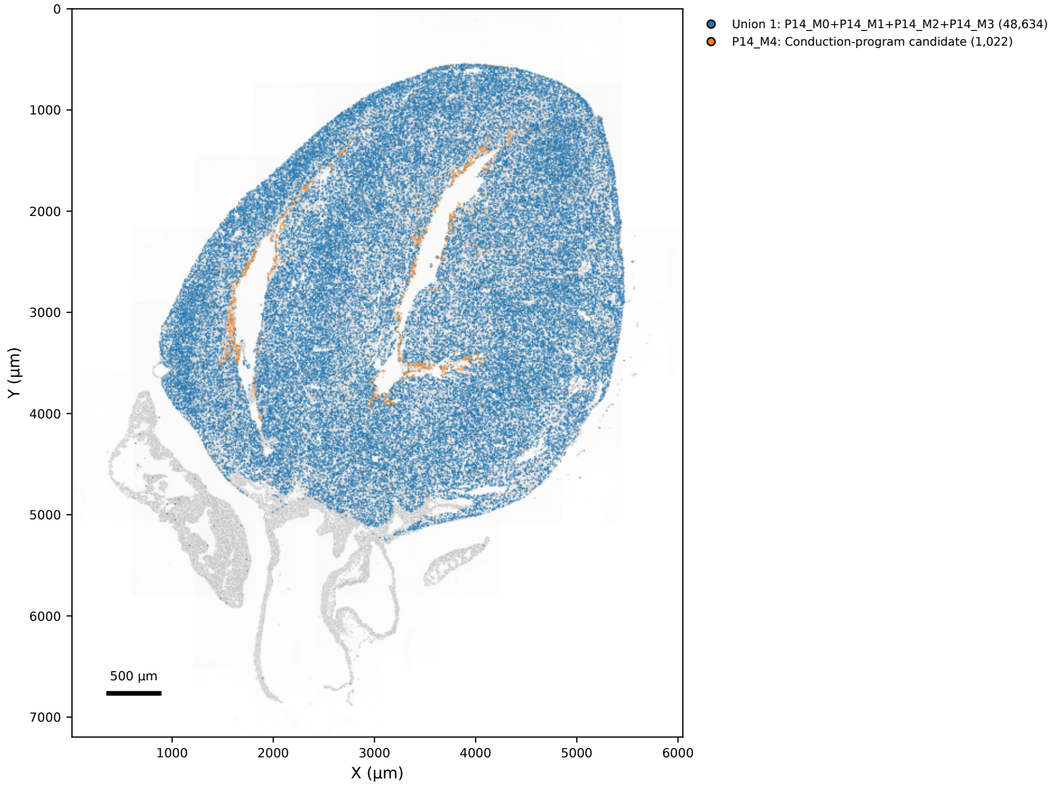
Finest exploratory proposals: every existing RNA group



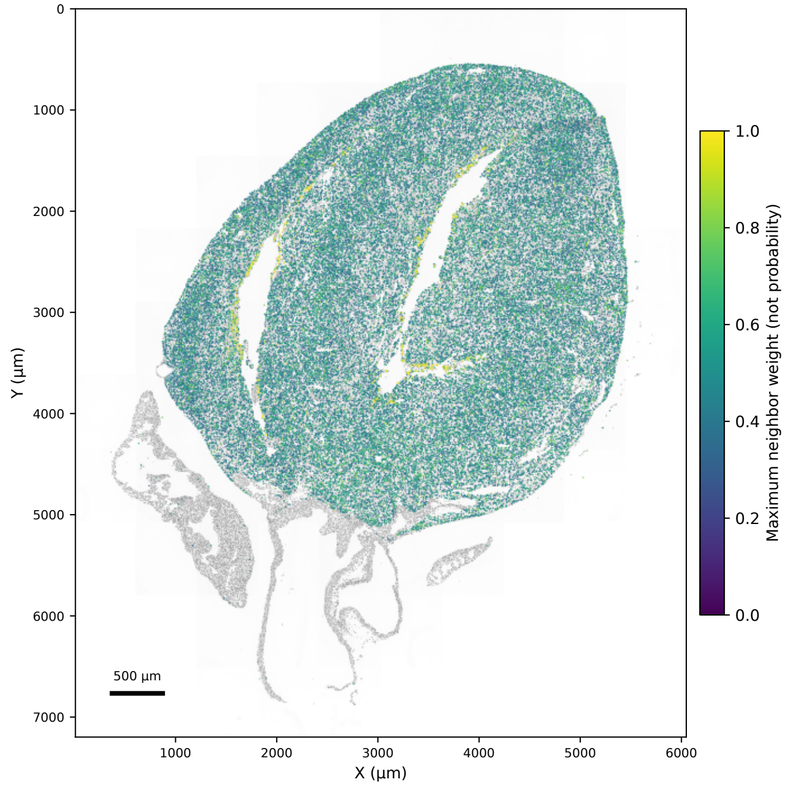
Fine candidates with either internal support check



Coarser cut selected on reserved RNA validation nuclei

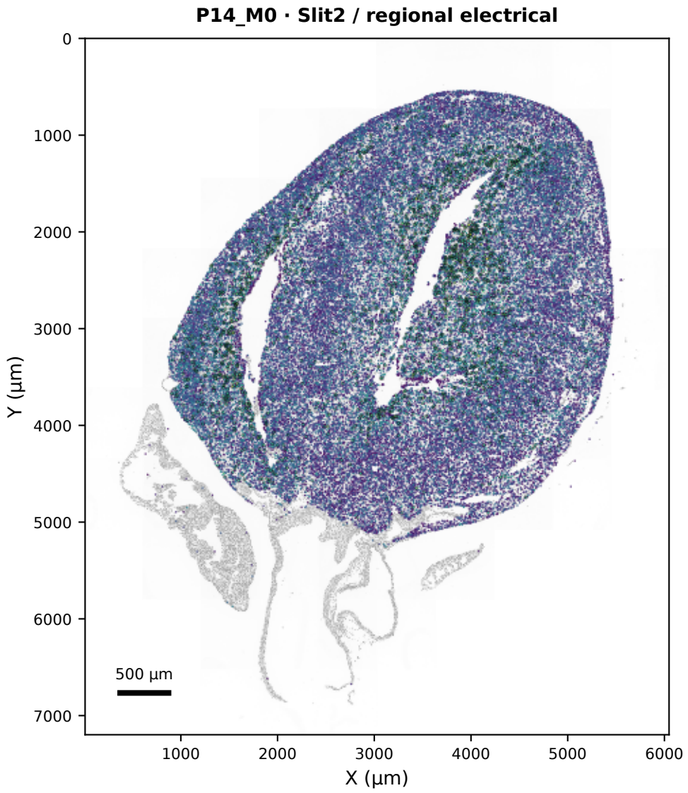


Fine-label weight concentration, without smoothing

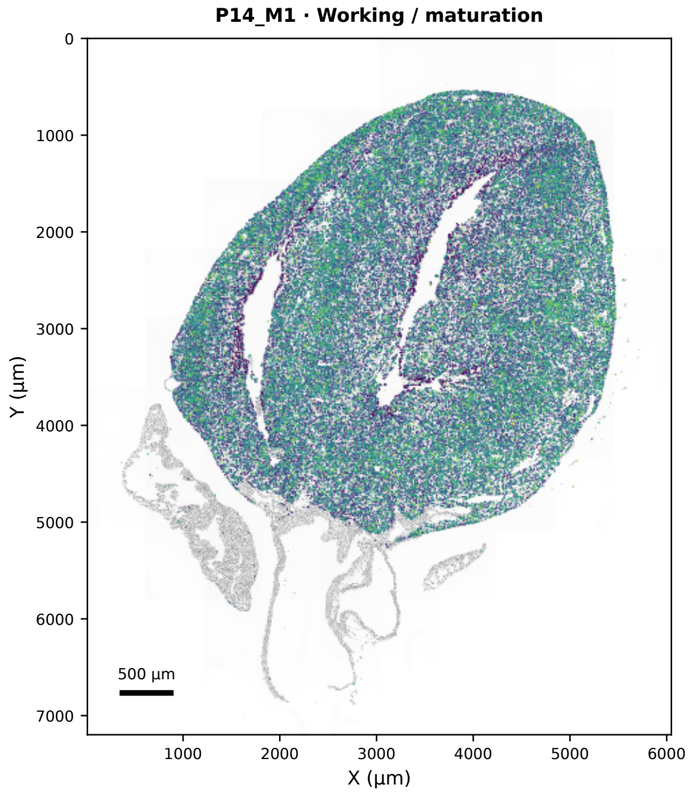


Top left is a permissive hypothesis map; top right adds internal RNA support, library/raw consistency and measured-marker co-detection.
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Conduction labels remain candidates. Kcnd2/Irx5 are unassayed; all working/electrical labels originate from the RNA reference.

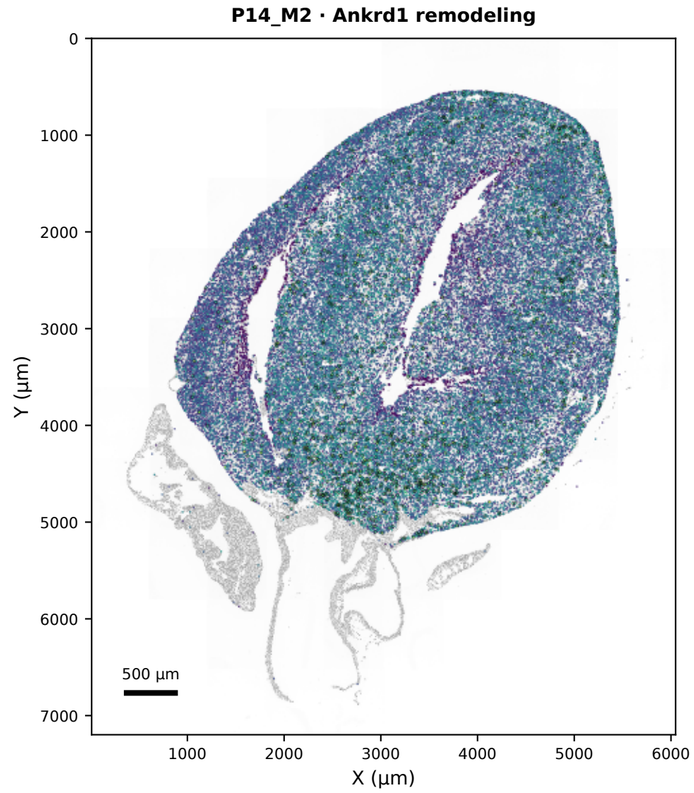
P14: individual spatial evidence maps for all 5 ventricular RNA groups



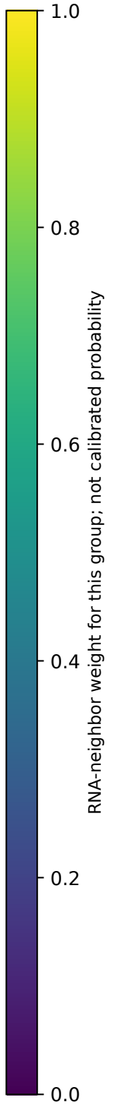
Proposed: 10,316; prioritized: 1,919
Measured supporting genes: Thsd4, Gm12381, Angpt1

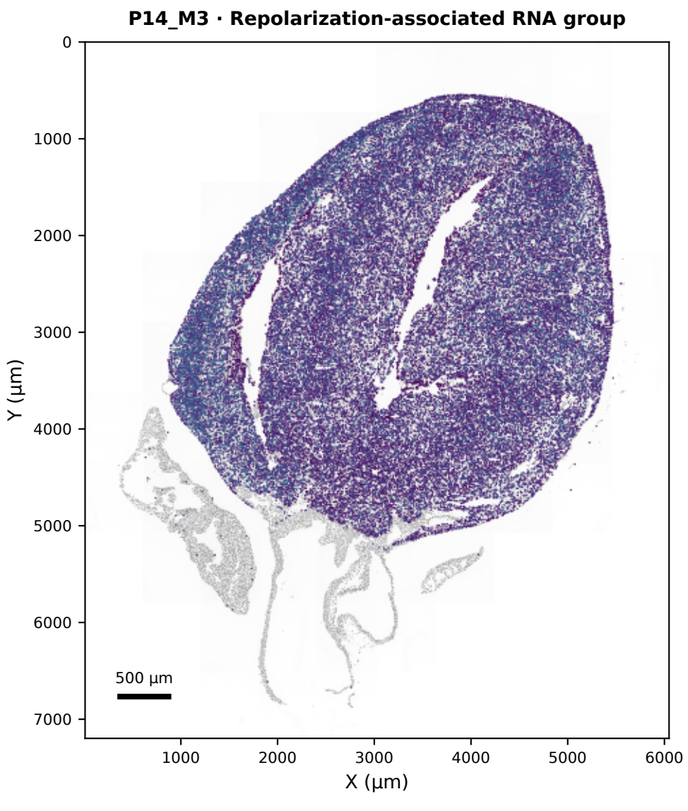


Proposed: 22,368; prioritized: 0
Measured supporting genes: none meeting marker rule

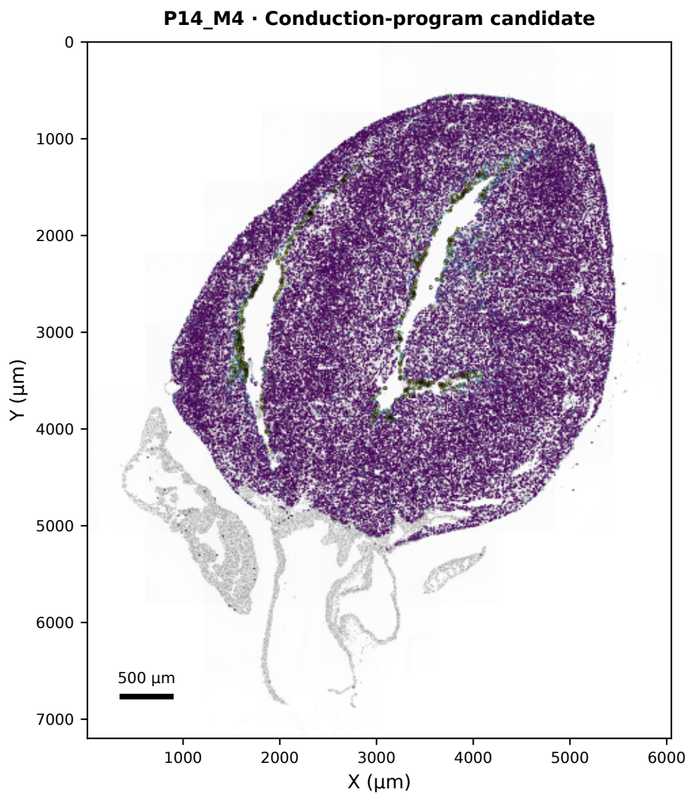


Proposed: 14,600; prioritized: 1,450
Measured supporting genes: Creb5, Tlr4, Ankrd1





Proposed: 1,025; prioritized: 0
Measured supporting genes: Acta2, Angpt1, Jam2

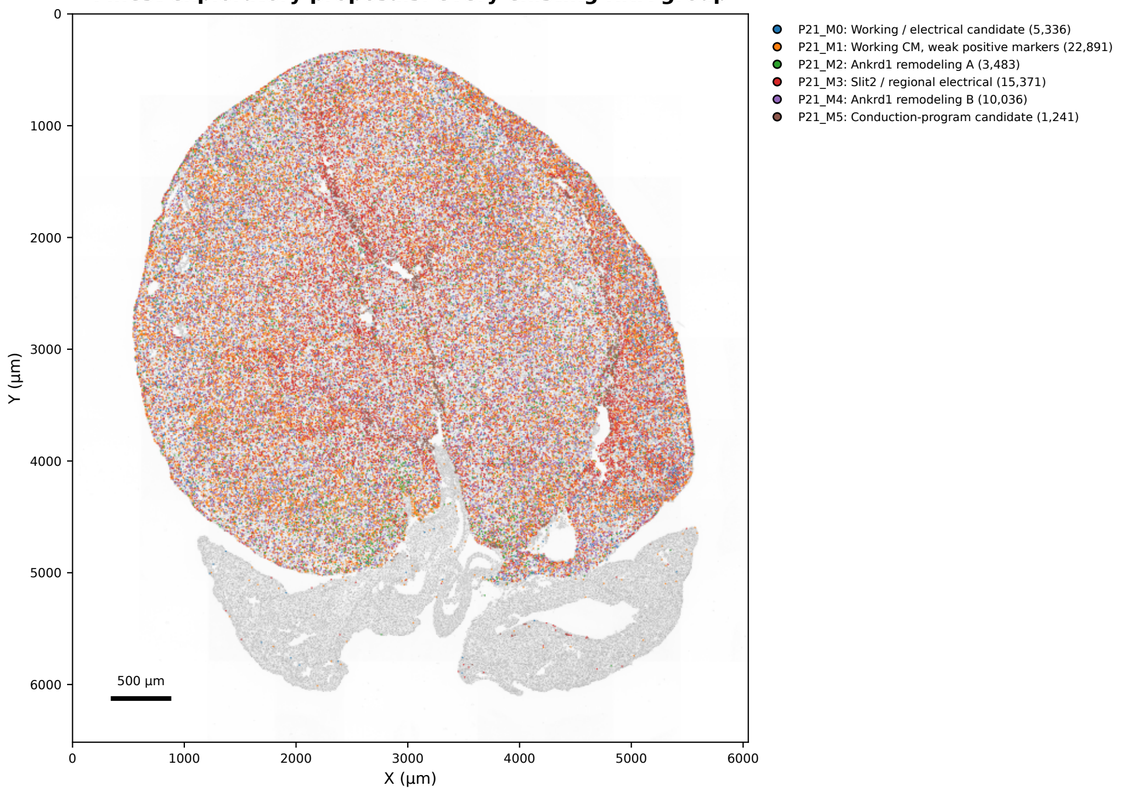


Proposed: 1,347; prioritized: 637
Measured supporting genes: Ephb1, Sema3a, Slit2

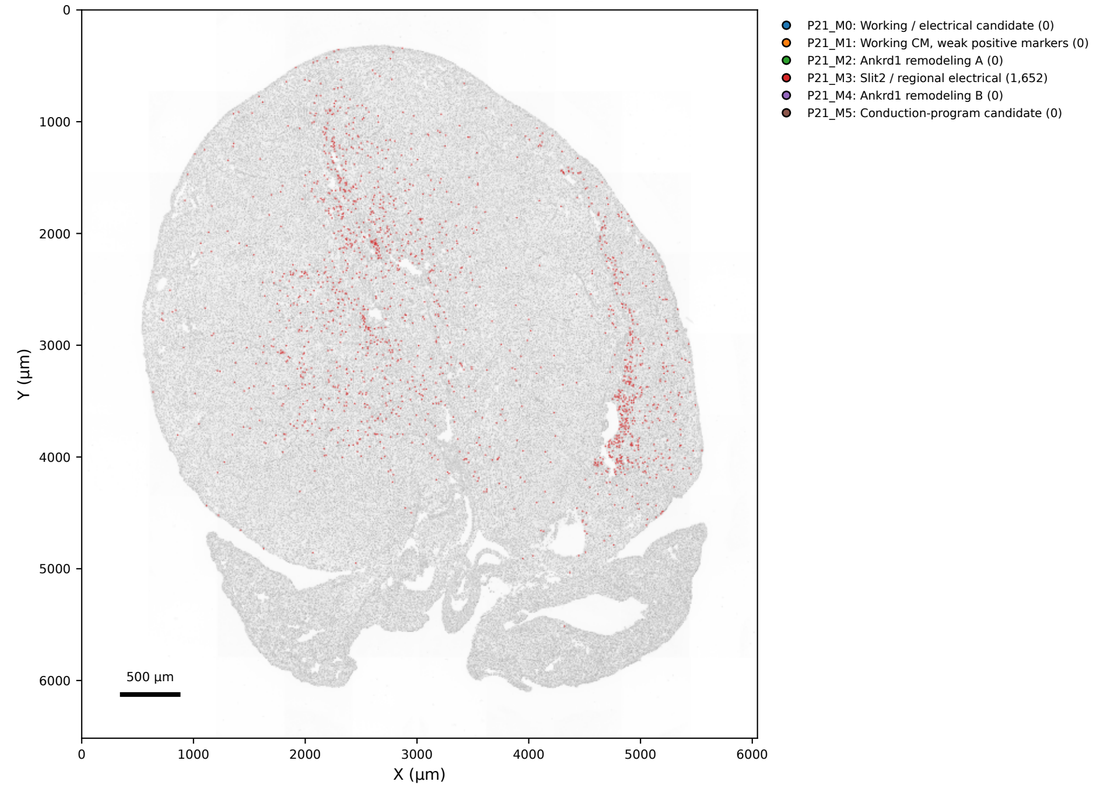
All original locations are retained. Colors show continuous group weight; black outlines mark prioritized candidate placements.
Measured markers corroborate labels but also contributed to matching. Low weights and absent outlines do not establish biological absence.

P21: pushing ventricular spatial mapping to 6 existing RNA groups

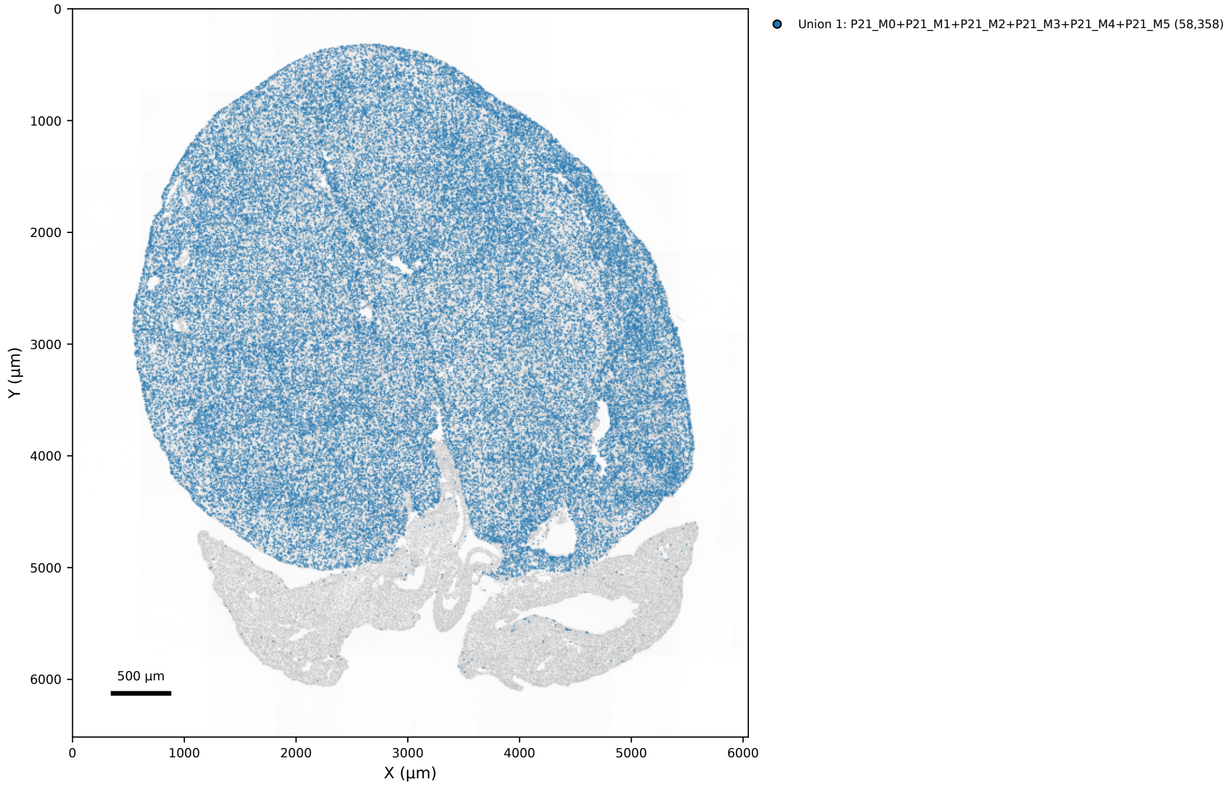
Finest exploratory proposals: every existing RNA group



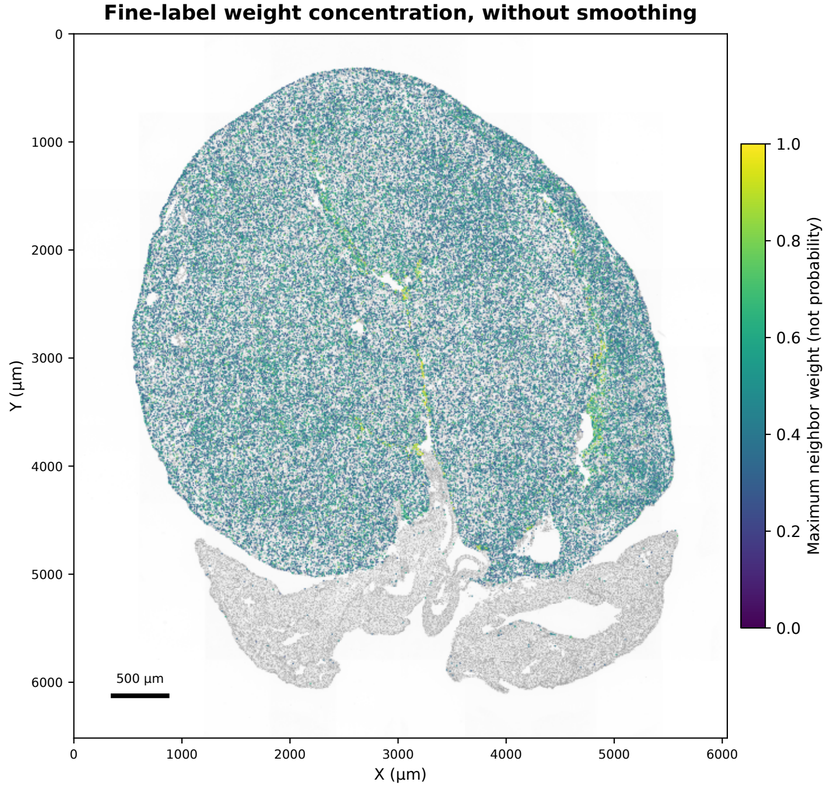
Fine candidates with either internal support check



Coarser cut selected on reserved RNA validation nuclei



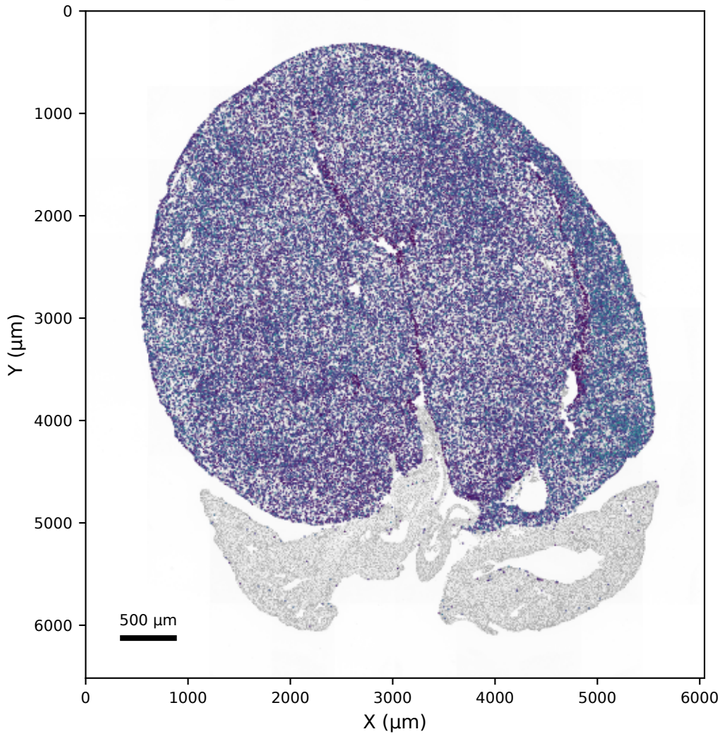
Fine-label weight concentration, without smoothing



Top left is a permissive hypothesis map; top right adds internal RNA support, library/raw consistency and measured-marker co-detection.
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Conduction labels remain candidates. Kcnd2/Irx5 are unassayed; all working/electrical labels originate from the RNA reference.

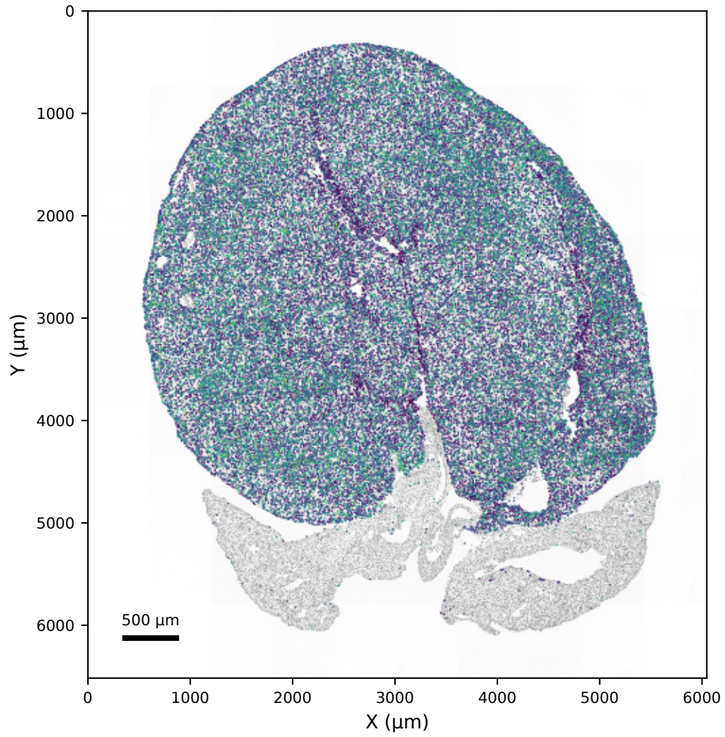
P21: individual spatial evidence maps for all 6 ventricular RNA groups

P21_M0 · Working / electrical candidate



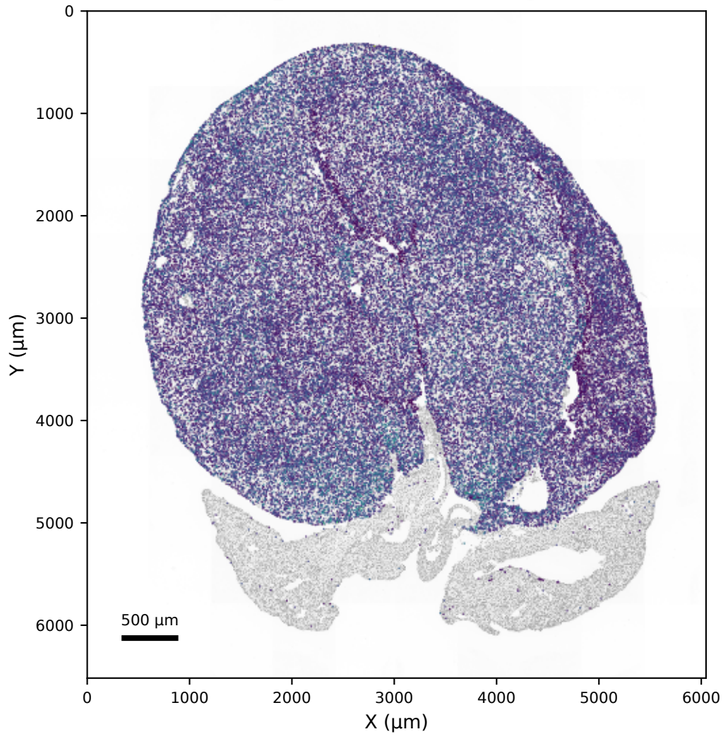
Proposed: 5,336; prioritized: 0
Measured supporting genes: Egf, Angpt1, Ppargc1a

P21_M1 · Working CM, weak positive markers



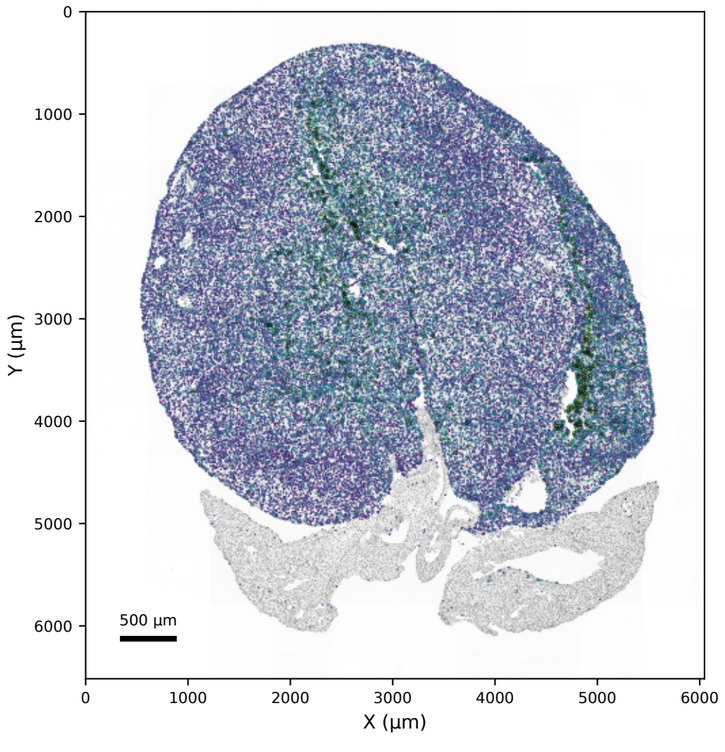
Proposed: 22,891; prioritized: 0
Measured supporting genes: none meeting marker rule

P21_M2 · Ankrd1 remodeling A



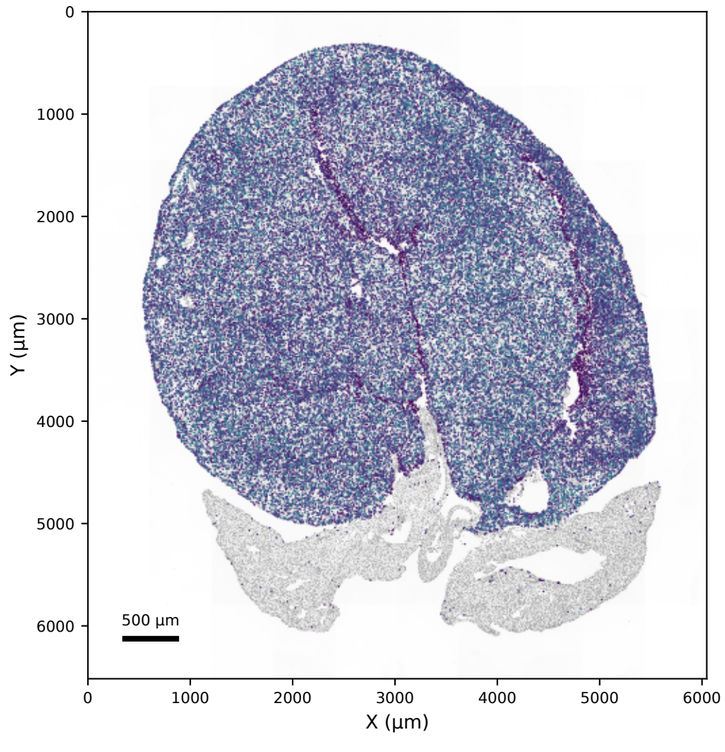
Proposed: 3,483; prioritized: 0
Measured supporting genes: Ankrd1, Creb5, Tlr4

P21_M3 · Slit2 / regional electrical



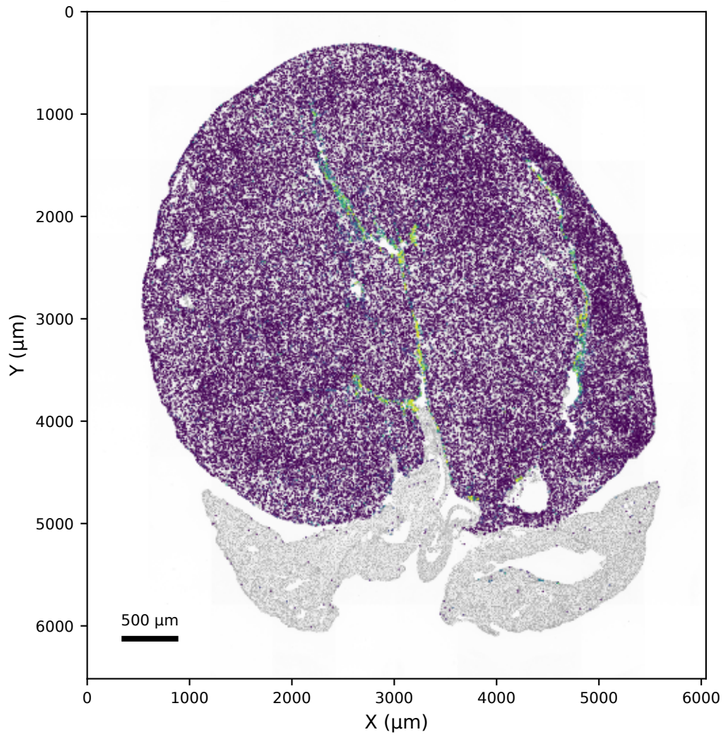
Proposed: 15,371; prioritized: 1,652
Measured supporting genes: Thsd4, Gm12381, Angpt1

P21_M4 · Ankrd1 remodeling B

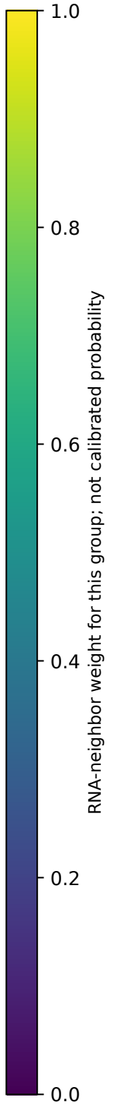


Proposed: 10,036; prioritized: 0
Measured supporting genes: Tlr4, Creb5, Ankrd1

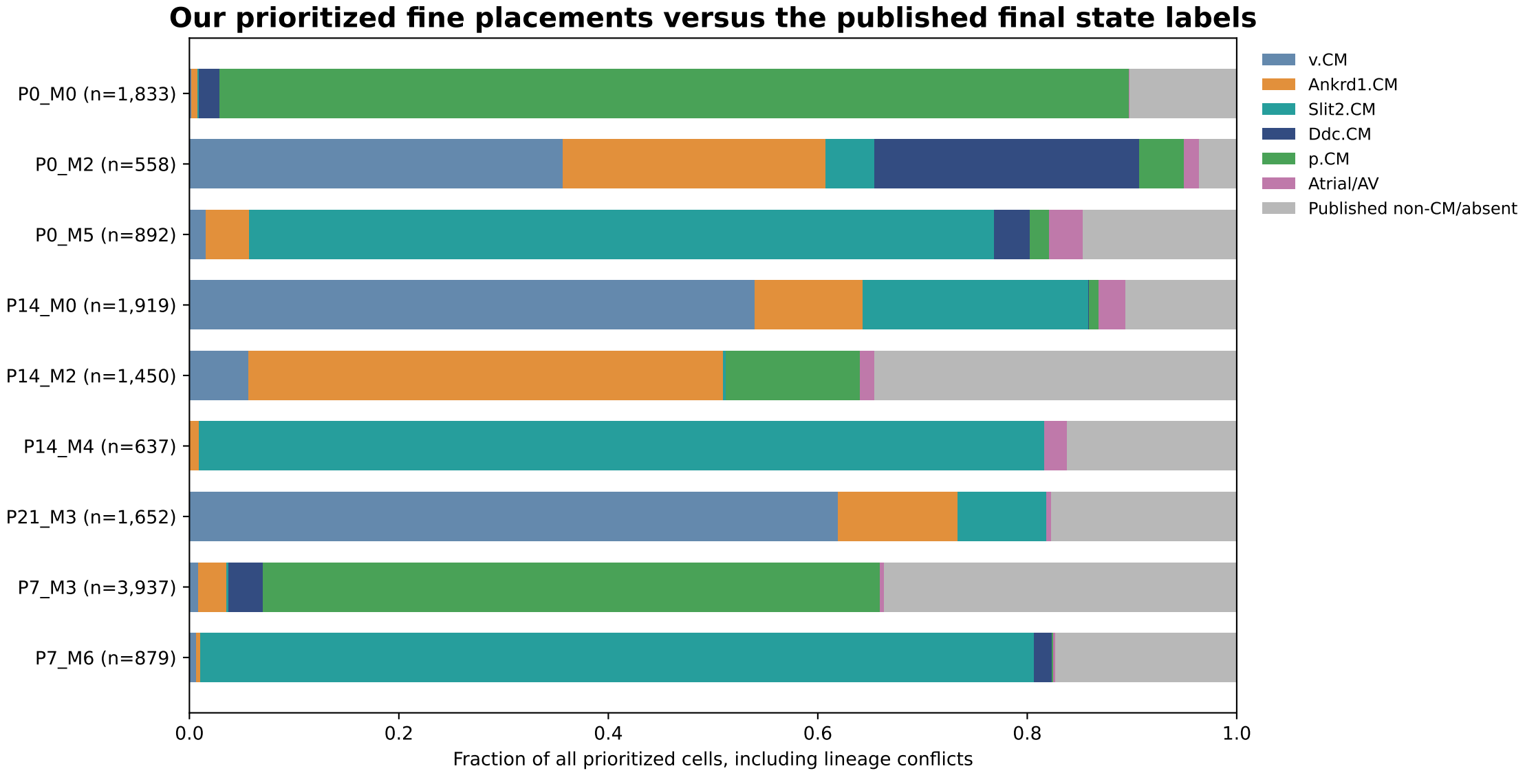
P21_M5 · Conduction-program candidate



Proposed: 1,241; prioritized: 0
Measured supporting genes: Slit2, Ephb1, Sema3a



All original locations are retained. Colors show continuous group weight; black outlines mark prioritized candidate placements.
Measured markers corroborate labels but also contributed to matching. Low weights and absent outlines do not establish biological absence.



These are annotation-overlap proportions, not accuracy estimates. Gray explicitly retains cells the authors labeled non-CM or omitted.
Author labels were not used for fitting or selecting our candidates. Agreement is strongest for Slit2/conduction-associated groups; lineage conflicts require review.